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Classification Lecture 9

Friday, 11 October 2024 6:08 AM

Regularization in Logistic Regression

↳ helps in reducing model overfitting. It adds a penalty term which encourages simpler models, leading to generalization on unseen data.

Lasso Regularization (L1 norm)

$$J(\beta) = \underbrace{-\log L(X;\beta)}_{\text{-log likelihood (cost function)}} + \underbrace{\lambda \|\beta\|_1}$$

max log likelihood
min - log likelihood

$$L(X;\beta) = \prod_{i=1}^{N} p(x_i)^{y_i} \cdot (1-p(x_i))^{1-y_i}$$

Updated Cost function:-

$$J(\beta) = -\left[\sum_{i=1}^{N} y_i \log p(x_i) + (1-y_i)\log(1-p(x_i))\right] + \lambda \|\beta\|_1$$

$$= -\left[\sum_{i=1}^{N} [-\log(1+e^{x_i\beta})] + \sum_{i=1}^{N} y_i x_i \beta\right] + \underbrace{\lambda \sum_{i=1}^{N} |\beta_i|}_{\text{L1 penalty}}$$

$$\frac{\partial}{\partial \beta_i}|\beta_i| = \text{sign}(\beta_i)$$

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$$\frac{\partial J(\beta)}{\partial \beta_0} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) + \lambda \cdot \text{sign}(\beta_0)$$

$$\frac{\partial J(\beta)}{\partial \beta_k} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) \cdot x_{ki} + \lambda \cdot \text{sign}(\beta_k)$$

Since this is a cost function, we can apply gradient descent to calculate β values.

Ridge Regularization (L_2 norm)

$$J(\beta) = \underbrace{-\log L(X;\beta)}_{\text{-log likelihood (cost function)}} + \underbrace{\lambda \|\beta\|_2^2}_{L_2 \text{ penalty}}$$

Updated Cost function :-

$$J(\beta) = -\left[\sum_{i=1}^{N} y_i \log p(x_i) + (1-y_i) \log(1-p(x_i))\right] + \lambda \|\beta\|_2^2$$

$$= -\left[\sum_{i=1}^{N} \left[-\log(1+e^{x_i\beta})\right] + \sum_{i=1}^{N} y_i x_i \beta\right] + \underbrace{\lambda \sum_{i=1}^{N} \beta_i^2}_{L_1 \text{ penalty}}$$

$$\frac{\partial}{\partial \beta_i} \beta_i^2 = 2\beta_i$$

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$$\frac{\partial J(\beta)}{\partial \beta_0} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) + 2\lambda\beta_0$$

$$\frac{\partial J(\beta)}{\partial \beta_k} = -\sum_{i=1}^{N} (y_i - \sigma(z_i)) \cdot x_{ki} + 2\lambda\beta_k$$

Multi-class classification using logistic Regression

$$P(y=k \mid \vec{x}) = \text{softmax}(z_k) = \frac{e^{x w_k}}{\sum_{i=1}^{K} e^{x w_i}} = \frac{e^{w_{0k} + w_{1k} \cdot x_1 + \cdots + w_{pk} \cdot x_p}}{\sum_{i=1}^{K} e^{w_{0i} + w_{1i} \cdot x_1 + \cdots + w_{pi} \cdot x_p}}$$

$$\# \text{ weights} = K \times (p+1)$$

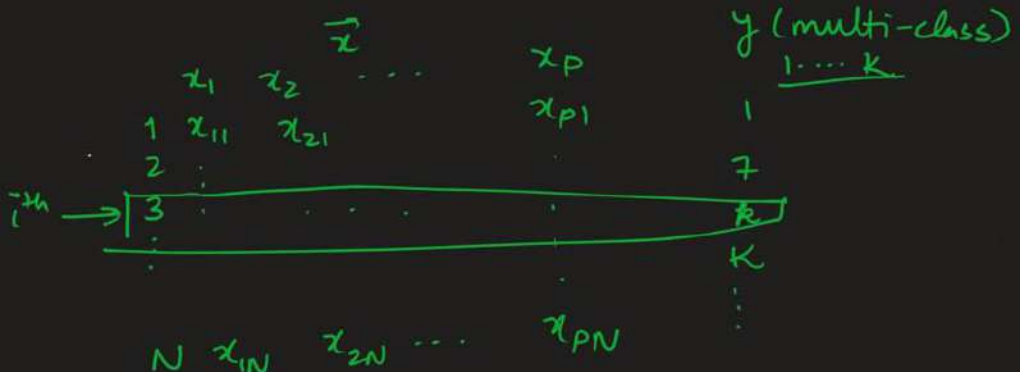
$$X \cdot W_k \quad (1 \times (p+1)) \quad ((p+1) \times 1)$$

Log-likelihood :- $$l(X, W) = \sum_{i=1}^{N} \sum_{j=1}^{K} 1\{y_i = k\} \log P\{y_i = k \mid x_i\}$$

indicator function, which is 1 if $y_i = k$, and 0 otherwise

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0 otherwise



		x_1	x_2	y (1/2/3)
x_1 →	1	3	1	1 ←
x_2	2	4	2	2 ←
x_3	3	7	5	3 ←

$$l = \log P(y=1 \mid x_1) + \log P(y=2 \mid x_2) + \log P(y=3 \mid x_3)$$

$$P(y=1 \mid x_1) = \frac{e^{x_1 \cdot w_1}}{e^{x_1 \cdot w_1} + e^{x_1 \cdot w_2} + e^{x_1 \cdot w_3}}$$

$$P(y=2 \mid x_2) = \frac{e^{x_2 \cdot w_2}}{e^{x_2 w_1} + e^{x_2 \cdot w_2} + e^{x_3 \cdot w_3}} \quad \ldots$$

> log-likelihood is a complex function of w_i vectors, we use gradient-ascent to compute w_i for every class.

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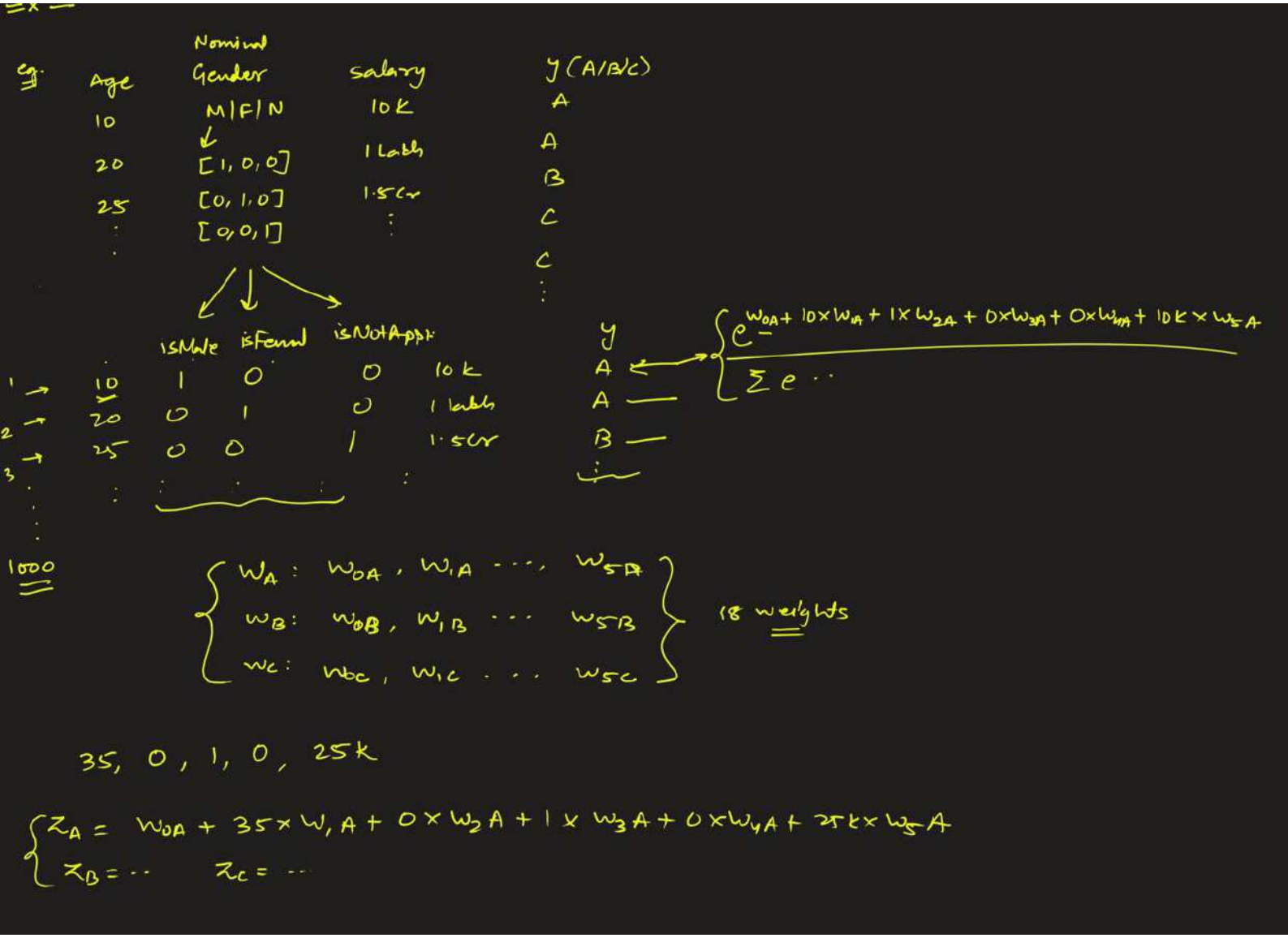
eg. We use logistic regression to compute the class of an unknown data point in 3-class classification problem. the logits for the classes A, B, C are $z_A = 3$, $z_B = 2.5$, $z_C = 4$. Find the probabilities of class A, B and C.

Sol. $P(A) = \text{Softmax}(z_A) = \frac{e^{z_A}}{e^{z_A} + e^{z_B} + e^{z_C}} = \frac{e^3}{e^3 + e^{2.5} + e^4} \approx 0.228$

$P(B) = \frac{e^{2.5}}{e^3 + e^{2.5} + e^4} \approx 0.138$, $P(C) \approx 0.638$

=x=

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Nominal
Eg.
Age
Gender
Salary
y (A/B/C)
10
20
25
M/F/N
10K
1 Lakh
1.5Cr
[1, 0, 0]
[0, 1, 0]
[0, 0, 1]
A
A
B
C
C
isMale
isFemale
isNotApplic
1
2
3
1000
y
A
A
B
$\frac{e^{w_{0A} + 10 \times w_{1A} + 1 \times w_{2A} + 0 \times w_{3A} + 0 \times w_{4A} + 10K \times w_{5A}}}{\Sigma e \ldots}$
W_A : $w_{0A}, w_{1A} \cdots w_{5A}$
W_B : $w_{0B}, w_{1B} \cdots w_{5B}$
W_C : $w_{0C}, w_{1C} \cdots w_{5C}$
18 weights
35, 0, 1, 0, 25k
$z_A = w_{0A} + 35 \times w_{1A} + 0 \times w_{2A} + 1 \times w_{3A} + 0 \times w_{4A} + 25k \times w_{5A}$
$z_B = \cdots$
$z_C = \cdots$